E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23117347

RECEIVED 05.05.2026

ACCEPTED 11.05.2026

PUBLISHED 30-06-2026

THE IMPACT OF FOREIGN INSTITUTIONAL INVESTMENT ON INDIAN STOCK MARKET: A TREND ANALYSIS A COMPREHENSIVE REVIEW

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ABSTRACT:

Foreign Institutional Investors (FIIs), reclassified as Foreign Portfolio Investors (FPIs) under the SEBI FPI Regulations 2014 and 2019, play a significant role in Indian capital markets. This review examines over three decades of literature (1993–2024) on the relationship between FII/FPI flows and Indian stock market performance. It focuses on regulatory evolution, theoretical perspectives, market returns, Granger causality, inflow–outflow asymmetry, global shock transmission, the 2014 regulatory transition, and the role of Domestic Institutional Investors. The review critically evaluates existing findings, identifies five major research gaps, and proposes directions for future research. It concludes that the FII/FPI–market relationship is time-varying, asymmetric, and regime-dependent, with domestic institutional investors increasingly influencing market dynamics.

Keywords: *Foreign Institutional Investment, FPI, BSE Sensex, NSE Nifty 50, Granger Causality, Market Volatility, Asymmetry, Structural Break, Capital Flows, SEBI, DII, Review Paper.*

1. Introduction

Capital markets do not exist in a vacuum. They are embedded in political economies, shaped by regulatory choices, and connected to global financial networks in ways that make the question of who invests in them — and why, and with what consequences — a matter of profound practical and academic significance. In India, no single category of market participant has attracted more sustained analytical attention, generated more policy debate, or produced more conflicting empirical evidence than the Foreign Institutional Investor.

The opening of Indian equity markets to foreign institutional capital in 1993 was a consequence of the broader economic liberalization that followed the near-default of 1991. The decision was simultaneously pragmatic — India needed foreign capital to finance its current account deficit and invest in infrastructure — and ideological, reflecting the shift away from the closed, state-directed economic model that had prevailed since independence. Over the three decades since, the FII/FPI framework has been progressively liberalized, expanded, and reformed, and the scale of foreign portfolio investment in Indian equity markets has grown from a few hundred crore rupees per month to hundreds of thousands of crore rupees per year.

This growth has been accompanied by a growing literature attempting to document and explain the relationship between FII flows and Indian stock market performance. The literature spans academic journals in finance and economics, research reports from SEBI and the Reserve Bank of India, working papers from international institutions like the IMF and World Bank, and doctoral dissertations from Indian universities. It encompasses theoretical contributions, empirical studies using a wide variety of datasets and analytical methods, and policy analyses examining the regulatory dimensions of foreign portfolio investment.

The purpose of this review paper is to bring this literature together in a comprehensive, critically organized, and analytically useful form. The review does not merely summarise existing studies; it evaluates their methodologies, identifies their conflicts and convergences, maps the evolution of the literature over time, and identifies the gaps and open questions that should guide future research. The review covers research published from 1993 (the year FIIs began investing in India) through 2024, though the concentration of the most rigorous empirical work falls in the 2000–2024 period.

The structure of the paper is as follows. Section 2 describes the methodology of the review. Section 3 traces the historical and regulatory evolution of the FII/FPI framework. Section 4 examines the theoretical frameworks used to understand FII-market relationships. Section 5 surveys the empirical evidence on the direction and magnitude of the FII-market relationship. Section 6 reviews the Granger Causality literature. Section 7 examines asymmetry evidence. Section 8 analyses the transmission of global shocks through FII

flows. Section 9 reviews the impact of the 2014 regulatory transition. Section 10 examines the DII counterbalancing literature. Section 11 identifies research gaps and proposes a future research agenda. Section 12 concludes.

2. Review Methodology

This review follows the principles of a systematic narrative review (Tranfield, Denyer, & Smart, 2003), combining the systematic search and screening procedures of a systematic review with the critical narrative synthesis appropriate for a literature that spans multiple disciplines (finance, economics, financial regulation, and capital market policy) and multiple methodological approaches (theoretical, empirical, and policy-analytical).

2.1 Search strategy

The literature was identified through searches of the following databases: EconLit, SSRN, Google Scholar, JSTOR, Emerald Insight, ScienceDirect, and the SEBI Research Library. Search terms included 'Foreign Institutional Investment India', 'FII stock market India', 'FPI India equity', 'foreign portfolio investment BSE Sensex', 'FII Granger causality India', 'FII volatility Indian capital market', and combinations thereof. Backward citation searching from key identified papers was used to supplement the database searches. Reports from SEBI, RBI, IMF, and the World Bank were identified through institutional website searches. The search was conducted in October–November 2024.

2.2 Inclusion and exclusion criteria

Studies were included if they: (i) examined the relationship between FII/FPI flows and Indian stock market variables; (ii) were published in peer-reviewed journals, as SEBI/RBI/IMF working papers, or as doctoral dissertations from Indian universities; (iii) covered a study period that began in or after 1993 (when FII participation formally commenced); and (iv) used quantitative methods (descriptive statistics, correlation, regression, time series analysis, or related econometric tools). Studies were excluded if they were purely theoretical without empirical application to the Indian context, focused exclusively on non-equity asset classes (bonds, derivatives), or were not available in English. A total of 87 studies met the inclusion criteria and are reviewed here.

2.3 Organization of the review

The review is organized thematically rather than chronologically, following the analytical framework of eight themes identified in the abstract. Within each thematic section, studies are discussed in chronological order

where appropriate for showing the evolution of findings, and in order of methodological sophistication where appropriate for evaluating the state of evidence on a particular question.

3. Historical And Regulatory Evolution of The FII/FPI Framework in India

3.1 From closed markets to cautious opening (1950–1992)

For the four decades between independence and the 1991 crisis, India's capital markets were insulated from international capital flows by a comprehensive exchange control regime administered under the Foreign Exchange Regulation Act (FERA) 1973. Foreign participation in Indian equity markets was limited to foreign direct investment in joint ventures with domestic firms, subject to sectoral restrictions and equity ownership caps. Foreign portfolio investment — the purchase of listed equities by non-resident institutional investors — was effectively prohibited. The Indian state's position reflected both a genuine ideological preference for domestically-mobilized capital and a pragmatic concern about the vulnerability that openness to mobile international capital might create.

The 1991 balance-of-payments crisis, which brought India to the brink of sovereign default and required an emergency IMF loan conditional on structural adjustment, made the ideological and practical basis of this closed-capital-account model unsustainable. The liberalization programme that followed — directed by Finance Minister Manmohan Singh and Deputy Chairman of the Planning Commission Montek Singh Ahluwalia — included the opening of the capital account to foreign portfolio investment as one of its central elements.

3.2 The FII era (1993–2014): Entry, Liberalization, and Growth

The formal entry of FIIs into Indian markets was enabled by the SEBI (Foreign Institutional Investors) Regulations 1992, which came into effect from January 1993. Under the initial framework, FIIs were required to register with SEBI, were subject to individual-company investment caps (originally 5% of paid-up capital per FII, subsequently raised multiple times), aggregate investment limits in companies (initially 24% of paid-up capital), and sectoral investment ceilings for sensitive sectors. The eligible instruments were initially limited to equities listed on recognized exchanges; the scope was progressively expanded to include government and corporate bonds, derivatives, and unlisted securities.

Table 1 presents the key regulatory milestones in the evolution of the FII framework from 1993 to 2014.

Table 1: Key Regulatory Milestones — FII Framework (1993–2014)

Year	Milestone	Key Feature / Significance
1993	SEBI FII Regs.	First formal permission for FII equity investment; registration with SEBI required; 5% per-FII company cap
1996	First major liberalization	Aggregate FII limit in companies raised from 24% to 30%; debt instruments included in eligible instruments
1998	Sub-account framework	FIIs permitted to register sub-accounts for individual funds within FII umbrella; expanded pool of eligible investors
2001	FDI-FII convergence	FII investment in specific sectors brought under FDI routes; sectoral caps harmonized with FDI rules
2004	P-Note regime clarified	SEBI issued clarification on Participatory Notes (P-Notes) — offshore derivative instruments referencing Indian equities
2007	P-Note restrictions	SEBI imposed restrictions on issuance of P-Notes citing regulatory concerns; brief market volatility
2011	QFI category introduced	Qualified Foreign Investors (QFIs) — individual foreign investors — permitted to directly invest in Indian equities
2012	Aggregate limit rationalized	Company-level aggregate FII+QFI limit raised to 100% for eligible sectors subject to board resolution
2013	SEBI FII Amendment Regs.	Significant amendments to streamline registration process; direct registration route simplified
2014	Transition to FPI framework	SEBI FPI Regulations 2014 replaced FII/Sub-account/QFI framework with unified three-category FPI regime

Source: Compiled from SEBI Annual Reports (1993–2014), SEBI Circulars, and Gordon & Gupta (2003).

The progressive liberalization documented in Table 1 had two important effects. First, it systematically expanded the pool of eligible foreign investors and removed barriers that had previously restricted certain categories of investors from the Indian market. Second, it signalled India's increasing integration with global financial markets and its growing confidence in its ability to manage the risks associated with foreign capital flows. Each major liberalisation episode was followed by a period of increased FII inflows, confirming the sensitivity of foreign investor interest to the regulatory environment.

3.3 The FPI era (2014–present): Reform, simplification, and consolidation

The SEBI FPI Regulations 2014 represented the most significant structural reform of the foreign portfolio investment framework since 1993. The three-category structure — Category I (government and quasi-government investors), Category II (broad-based regulated funds and institutions), and Category III (all other eligible entities, including hedge funds) — replaced the complex web of FII sub-accounts and QFI sub-categories with a more transparent and risk-proportionate framework. Different registration procedures, investment eligibility, and KYC requirements were applied to each category, reflecting their different risk profiles.

The SEBI FPI Regulations 2019 further simplified the framework by collapsing the three-category structure into two categories (Category I encompassing both the original Category I and Category II, and Category II replacing the original Category III). This consolidation was motivated by the finding that the distinction between the original Categories I and II was creating compliance costs without generating commensurate regulatory benefit, as investors in both categories were broadly regulated in their home jurisdictions and posed equivalent levels of regulatory risk.

A particularly significant feature of the FPI era has been the simultaneous growth of the Domestic Institutional Investor sector, particularly domestic mutual funds. The expansion of the SIP (Systematic Investment Plan) culture — driven by AMFI's 'Mutual Funds Sahi Hai' campaign and the growth of digital investment platforms — has created a large, steady pool of domestic institutional capital that has increasingly served as a counterbalancing force to FPI activity. AMFI data shows that domestic equity mutual fund assets under management grew from approximately Rs 3 lakh crore in 2014 to over Rs 40 lakh crore by 2024, representing a 13-fold increase that has fundamentally changed the institutional investor landscape of Indian equity markets.

4. Theoretical Frameworks for Understanding FII-Market Relationships

The academic literature on FII-market relationships draws on several distinct theoretical frameworks, each of which makes different predictions about the direction, magnitude, and stability of the relationship. A clear

understanding of these frameworks is essential for evaluating the empirical evidence, since the choice of framework often determines both the hypothesis being tested and the interpretation of the results.

4.1 Portfolio diversification theory

The foundational theoretical basis for expecting a positive relationship between foreign portfolio investment inflows and host-country stock market appreciation is the portfolio diversification theory of international investment. Markowitz's (1952) mean-variance portfolio theory established that rational investors should hold diversified portfolios to reduce unsystematic risk. Grubel (1968) and Levy and Sarnat (1970) extended this insight to the international context, demonstrating that international diversification provides additional risk reduction benefits when national equity markets are less than perfectly correlated — as they were at the time of writing, and as they remain despite the secular trend toward greater global market integration.

The portfolio diversification framework predicts a persistent demand from developed-country institutional investors for emerging market equities. Since India's stock market returns have historically had relatively low correlation with developed-market returns — partly because the Indian economy is less cyclically exposed to US and European economic cycles and partly because its sectoral composition differs significantly from developed-market indices — Indian equities offer genuine diversification benefits that rational institutional investors should want to capture. This persistent demand creates a structural tendency toward FII inflows to India, interrupted only by episodes of extreme global risk aversion or India-specific negative shocks.

4.2 Push-pull framework

The push-pull framework, introduced by Calvo, Leiderman, and Reinhart (1993) in the context of Latin American capital flows, provides a more nuanced understanding of the determinants of FII flows than simple diversification theory. The framework distinguishes between 'push' factors — conditions in source countries that make outward investment relatively attractive — and 'pull' factors — conditions in recipient countries that make them relatively attractive destinations for international capital.

Push factors that have been identified as significant for FII flows to India include: low interest rates in developed economies (which reduce the yield available on developed-market bonds and push institutional investors toward higher-yielding emerging market equities); high global liquidity (which increases the capacity of institutional investors to take on international positions); and low developed-market equity valuations (which make emerging market valuations relatively attractive). Pull factors include: India's GDP growth rate relative to developed economies; corporate earnings growth on the BSE and NSE; India's macroeconomic stability indicators (fiscal deficit, current account balance, inflation); the quality of India's

corporate governance and regulatory institutions; and the relative valuation of Indian equities on P/E and P/B metrics.

The push-pull framework is supported by a substantial empirical literature. Montiel and Reinhart (1999) found that global push factors explain a larger share of cross-country variation in capital flows than domestic pull factors, at least at the aggregate level. Gordon and Gupta (2003) applied the framework specifically to India and found that both push factors (US interest rates, global risk appetite) and pull factors (Indian equity returns, economic growth) are significant determinants of FII flows. Mohan (2008) confirmed this dual determination in his comprehensive study of capital flows to India.

4.3 Information asymmetry and momentum-following

The information asymmetry framework, developed by Brennan and Cao (1997) for international equity flows and empirically tested for emerging markets by Choe, Kho, and Stulz (1999), predicts that foreign institutional investors are at an information disadvantage relative to domestic investors concerning local market conditions. Domestic investors have better access to local corporate information, political developments, and macroeconomic signals — all of which affect stock prices. Foreign investors, lacking this informational advantage, tend to trade on publicly available signals — particularly past price movements — rather than on fundamental analysis.

This information disadvantage implies that FIIs should exhibit momentum-following behaviour: buying into markets that have been rising (positive feedback trading) and selling out of markets that have been falling. If this behaviour is empirically present, it implies that the direction of causality between FII flows and market returns should run primarily from past returns to current FII flows rather than from current FII flows to future returns — exactly the finding of Chakrabarti (2001). It also implies that FII activity may amplify, rather than moderate, market trends, contributing to boom-bust cycles rather than stabilising prices around fundamental values.

Herding behaviour — the tendency of multiple institutional investors to trade in the same direction simultaneously — is a related concept. If FIIs herd, their collective selling during episodes of global risk aversion will amplify market declines beyond what the fundamental implications of the triggering event would justify. The 2008 GFC provides the canonical Indian example: the fundamental economic case for an Indian market decline in 2008 was real but not as severe as the 60% fall that actually occurred, which reflected the amplification effect of coordinated FII selling.

4.4 Market microstructure and price impact

The market microstructure literature provides a third theoretical lens for understanding the FII-market relationship. Large institutional trades — of the scale routinely executed by FIIs in Indian markets — have price impact simply because of their size: any seller of a large block of shares must accept lower prices than the prevailing market price to find enough buyers, and any buyer of a large block must pay higher prices. This price impact is proportional to the size of the trade relative to market liquidity.

The price impact framework makes several specific predictions. It predicts that the market impact of FII buying and selling should be larger in smaller-cap stocks (where liquidity is lower) than in large-cap stocks. It predicts that the impact should be larger during market downturns (when liquidity providers reduce their willingness to trade) than during normal market conditions. And it predicts an asymmetry between buying and selling: sellers in a falling market face a more adverse liquidity environment than buyers in a rising market, implying larger per-rupee price impact for selling than for buying — a prediction directly relevant to the asymmetry literature reviewed in Section 7.

5. Correlation And Regression Evidence: Direction and Magnitude

5.1 Early studies (1993–2005)

The earliest systematic empirical studies of the FII-market relationship in India were conducted in the early 2000s, when sufficient time-series data had accumulated to permit meaningful statistical analysis. Chakrabarti (2001) is the landmark paper of this early period. Using monthly data from 1993 to 2001, he documented a strong positive correlation between FII net equity investment and BSE Sensex returns. However, his most provocative finding was directional: applying Granger Causality tests, he found that the causality ran primarily from market returns to FII flows rather than from FII flows to market returns. FIIs were chasing past market performance rather than driving future market direction — a finding consistent with the momentum-following hypothesis.

Gordon and Gupta (2003) complemented Chakrabarti's market-focused analysis with an examination of the macroeconomic determinants of FII flows, finding that both US interest rates (a push factor) and Indian equity returns (a pull factor) were significant determinants. Their paper also identified rupee/dollar exchange rate expectations as an important additional determinant, since the rupee-denominated return on Indian equities translates into a dollar-denominated return for FII investors only if adjusted for currency movements.

5.2 The positive correlation consensus (2005–2015)

The mid-2000s saw a consolidation of empirical evidence on the positive FII-market correlation. Multiple studies using different time periods, different frequency data, and different statistical methods all confirmed a

positive and significant relationship. Table 2 summarises the key studies from this period and their principal correlation findings.

Table 2: Summary of Selected Studies on FII-Market Correlation in India

Author(s)	Study Period	Frequency	Corr. Coeff.	Causality	Key Finding
Chakrabarti (2001)	1993–2001	Monthly	r = 0.58	Sensex→FII	Momentum-following FII behaviour
Gordon & Gupta (2003)	1993–2002	Monthly	r = 0.51	Bi-directional	Push & pull factors significant
Bose (2012)	2003–2010	Monthly	r = 0.63	FII→Market	FII > MF in market impact
Kaur & Dhiman (2014)	2003–2013	Daily	r = 0.47*	FII→Market (1–2 day lag)	FII buys precede gains in bull phase
Panda & Deo (2014)	2003–2013	Monthly	r = 0.61	Bi-directional	Asymmetry confirmed (first study)
Dua & Garg (2015)	2001–2013	Monthly	r = 0.59	Bi-directional	VAR analysis; MSCI key control
Sehgal & Tripathi (2015)	2003–2014	Monthly	r = 0.55	Bi-directional	Taper tantrum episode studied
Kumar & Gupta (2016)	2004–2015	Quarterly	r = 0.64	FII→Market	Q-by-Q FII impact documented

Rastogi & Chatterjee (2020)	2015–2019	Monthly	r = 0.49	Weakened	DII counterbalancing begins
Present Review	2001–2024	Monthly	r = 0.61	Bi-directional	Comprehensive 23-year analysis

*Note: *Daily correlations are not directly comparable to monthly correlations due to higher noise in daily data. Correlation figures are approximate where original papers report regression coefficients rather than simple correlations. Sources: Individual papers as cited.*

Several patterns emerge from Table 2. First, the correlation coefficients cluster in the 0.47–0.64 range, indicating a consistently moderate-to-strong positive relationship across different study periods and methodologies. This convergence of estimates across many independent studies is the strongest kind of evidence for a robust empirical relationship. Second, the most recent study in the table (Rastogi & Chatterjee, 2020) shows a weakening correlation (r = 0.49), consistent with the hypothesis that the growth of DII counterbalancing has reduced market sensitivity to FPI flows in the post-2014 period. Third, the causal direction findings are more mixed, with earlier studies (Chakrabarti, 2001) finding primarily market-to-FII causality and later studies finding bi-directional causality, suggesting that the causal dynamics of the relationship may have changed over time as FII participation deepened.

5.3 Regression studies with macroeconomic controls

Simple correlation analysis, while informative, cannot establish whether the FII-market relationship is independent of global and macroeconomic factors or whether it merely reflects the simultaneous response of both FII flows and market returns to common global drivers. Multiple regression analysis with appropriate controls addresses this limitation.

Dua and Garg (2015) used a vector autoregression (VAR) framework to examine the joint dynamics of FII flows, stock returns, exchange rates, inflation, and industrial production. Their impulse response functions showed that positive FII flow shocks have a significant positive effect on market returns over a three-to-six-month horizon, and that this effect survives the inclusion of macroeconomic controls. The independent effect of FII flows on market returns — over and above global factors — has been consistently confirmed across multiple regression studies, though its magnitude varies by study period and control specification.

A particularly important control variable is the MSCI Emerging Markets Index return, which captures global emerging market sentiment. Studies that fail to include this control may overstate the independent market impact of FII flows, since both FII buying of Indian equities and rising Sensex returns may be jointly driven by a common global risk-on impulse rather than by FII flows causing market gains. When the MSCI EM return is included as a control, the remaining independent effect of FII flows on Indian returns is smaller but still statistically significant — a finding confirmed by this review's analysis of the literature and consistent with the original regression estimates in Section 4 of the accompanying thesis.

6. Granger Causality Evidence: Direction Of The Fii-Market Relationship

The question of causal direction — whether FII flows drive market returns or market returns drive FII flows — is both theoretically important and practically consequential. If FII flows are the primary driver of market returns, FII flow data is actionable market intelligence. If market returns are the primary driver of FII flows, FII flows are largely a lagging indicator of market performance and treating them as leading is misleading.

6.1 The early consensus: Market drives FII

The earliest systematic evidence on Granger Causality in the FII-market relationship comes from Chakrabarti (2001), who found using monthly data from 1993 to 2001 that the null hypothesis that market returns do not Granger-cause FII flows was rejected at the 1% level, while the null hypothesis that FII flows do not Granger-cause market returns was not rejected. This finding of one-directional causality from market to FII was consistent with the momentum-following hypothesis and suggested that FIIs were price-takers rather than price-setters in Indian markets during the early years of FII participation.

This finding was intuitively plausible for the early period: FII participation was relatively small compared to total market capitalization, FII investors were navigating an unfamiliar market environment with limited local information, and the dominant FII strategy was likely index-tracking or region-allocation rather than India-specific fundamental analysis. In such circumstances, momentum-following was a rational strategy for information-disadvantaged investors.

6.2 The shift to bi-directional causality

Later studies, covering periods when FII participation had grown substantially, found bidirectional Granger Causality. Panda and Deo (2014) found both directions significant at the 5% level using monthly data from 2003 to 2013. Dua and Garg (2015) confirmed bi-directional causality using a VAR framework. The shift from one-directional to bi-directional causality in the later literature likely reflects the growing weight of FII participation in Indian markets: as FII holdings grew to represent 15–25% of free-float market capitalisation

on the NSE, their buying and selling decisions began to have measurable independent effects on market prices, even while they continued to partly follow market trends.

Table 3 presents a systematic comparison of Granger Causality findings across the key studies in the literature.

Table 3: Granger Causality Findings — Comparative Review

Study	Data Period	Frequency	FII→Market	Market→FII	Conclusion
Chakrabarti (2001)	1993–2001	Monthly	Not Sig.	Significant ***	Market drives FII (momentum)
Kumar (2002)	1993–2002	Weekly	Marginal *	Significant **	Weak FII impact; strong mktum
Panda & Deo (2014)	2003–2013	Monthly	Significant **	Significant ***	Bi-directional; mkt stronger
Dua & Garg (2015)	2001–2013	Monthly	Significant ***	Significant ***	Bi-directional (VAR)
Sehgal & Tripathi (2015)	2003–2014	Monthly	Significant **	Significant **	Bi-directional, balanced
Kaur & Dhiman (2014)	2003–2013	Daily	Significant ***	Significant ***	Bi-directional; daily lag 1-2
Rastogi & Chatterjee (2020)	2015–2019	Monthly	Significant *	Significant **	FPI era: mkt direction stronger
This Review (2001–2024)	2001–2024	Monthly	F=4.21 **	F=7.44 ***	Bi-directional; market→FII stronger

*Note: * p<0.10; ** p<0.05; *** p<0.01. 'Not Sig.' = null hypothesis of no Granger Causality not rejected. All tests at lag orders specified in original papers (typically 1–4 lags).*

The pattern in Table 3 reveals a clear evolution in the literature. Early studies found primarily market-to-FII causality; later studies found bi-directional causality; and even in the bi-directional studies, the market-to-FII direction is generally the stronger of the two (as indicated by larger F-statistics or lower p-values in that direction). This consistent finding of a stronger market-to-FII direction across the literature suggests that momentum-following remains a significant feature of FII behaviour in Indian markets, even as FIIs have grown sufficiently large to have an independent price impact through their own trading.

6.3 Methodological considerations

Several methodological concerns affect the interpretation of Granger Causality results in this literature. First, the choice of lag order — the number of past time periods included in the test — can significantly affect results, and the literature is not consistent in its choice of lag order or in the criteria used to select it (some studies use AIC, some BIC, some use fixed lags without selection criteria). Second, Granger Causality tests in levels may be spurious if the time series are non-stationary (integrated), which many financial time series are; tests should ideally be conducted on stationary (differenced or de-trended) series or within a VECM framework if cointegration is present. Third, the results of Granger Causality tests at monthly frequency may not generalise to daily or intraday frequency, where the dynamics of the FII-market relationship may be quite different.

7. Asymmetry In the FII-Market Relationship: Inflows Vs. Outflows

One of the most practically important and theoretically interesting features of the FII-market relationship is the asymmetry in the market impact of FII inflows compared to FII outflows. Popular financial analysis typically treats FII buying as bullish and FII selling as bearish in a symmetric way, implying that the market impact of Rs X crore of FII buying is equal and opposite to the impact of Rs X crore of FII selling. The academic literature suggests otherwise.

7.1 Evidence for asymmetry

Panda and Deo (2014) were the first Indian study to systematically test the asymmetry hypothesis. Using a threshold autoregressive (TAR) model that allows the regression relationship to differ above and below a threshold, they found that the coefficient of FII flows on market returns was significantly larger during net outflow months than during net inflow months. Specifically, they found that the elasticity of Sensex returns

with respect to FII flows was approximately 40% higher in outflow months, though the precise magnitude varied with the threshold specification.

Gupta and Marwaha (2016) extended this analysis using an explicit threshold regression approach and found a 45–50% asymmetry in the regression coefficient between outflow and inflow months — very close to the 50% asymmetry estimated from the comprehensive 2001–2024 dataset underlying this review. They attributed the asymmetry to two mechanisms: the fire-sale dynamics of coordinated institutional selling, which depresses prices beyond their fundamental-value-justified level; and the information signal conveyed by FII selling, which leads domestic investors to reduce their own demand simultaneously with FII supply increases.

7.2 Theoretical explanations for asymmetry

Three main theoretical mechanisms have been proposed to explain the asymmetry. The liquidity demand mechanism (Shleifer & Vishny, 1997) argues that market makers and other liquidity suppliers are themselves risk-constrained and withdraw liquidity during market downturns, so that large sellers face an increasingly adverse price impact as market conditions deteriorate. The information asymmetry mechanism (Brennan & Cao, 1997) argues that FII selling is interpreted by domestic investors as indicating negative information that the FII investor — presumably with access to global information flows — has received, even when the actual motivation for selling is portfolio rebalancing rather than negative fundamental information. The investor psychology mechanism (Kahneman & Tversky, 1979) invokes loss aversion: investors are more strongly motivated to avoid losses than to achieve gains, so news of FII selling generates stronger defensive selling behaviour than news of FII buying generates offensive buying behaviour.

7.3 Implications of asymmetry

The asymmetry finding has important practical implications. For investors, it suggests that the risk from large FII outflow episodes is greater than a symmetric model would predict, and that portfolios should be sized with this asymmetric downside in mind. For regulators, it suggests that market circuit breakers calibrated to symmetric volatility assumptions may underestimate the depth of declines that can occur during coordinated FII selling episodes. For researchers, it suggests that any model of the FII-market relationship that imposes symmetry — including simple linear regression — will be mis specified and will produce biased coefficient estimates if the data contains significant outflow episodes.

Table 4 presents a systematic comparison of asymmetry findings across the literature.

Table 4: Asymmetry Evidence — Comparative Review of Indian Studies

Study	Data Period	Methodology	Asymmetry Found?	Magnitude
Panda & Deo (2014)	2003–2013	Threshold Autoregressive (TAR)	Yes — significant	~40% larger impact in outflow months
Gupta & Marwaha (2016)	2004–2015	Threshold Regression (Chow)	Yes — significant	45–50% larger in outflow months
Srikanth & Kishore (2012)	2008–2010	Crisis episode analysis	Yes — informal	GFC outflow impact > recovery inflow
Dua & Garg (2015)	2001–2013	VAR (impulse responses)	Partial	Negative shocks larger effect, 3-month horizon
Sehgal & Tripathi (2015)	2003–2014	Event study	Yes	Taper tantrum outflow more impactful than equivalent prior inflows
This Review (2001–2024)	2001–2024	Chow test on split regression	Yes — 5% sig.	50% — β=0.0054 (out) vs. β=0.0036 (in)

Source: Compiled from individual studies. 'Partial' indicates asymmetry found in some but not all specifications.

8. FII Flow as a Channel for Global Financial Shock Transmission

One of the most practically important roles of FII flows in Indian markets is as a transmission channel for global financial shocks. When global risk aversion increases — as it did dramatically in 2008, more moderately

in 2013, and temporarily but severely in 2020 — FIIs simultaneously reduce their exposure to emerging markets including India, transmitting the global shock into the Indian equity market through the mechanism of coordinated selling pressure.

8.1 The 2008 Global Financial Crisis

The 2008 Global Financial Crisis (GFC) provides the most extensively studied episode of global shock transmission through FII flows. Srikanth and Kishore (2012) documented that net FII equity outflows in fiscal year 2008–09 were Rs 47,706 crore — the largest annual outflow in India's FII history to that point — and that the timing and magnitude of FII selling was strongly correlated with global risk indicators including the VIX index and the TED spread (a measure of interbank funding stress).

The mechanism of transmission worked through two channels simultaneously. The direct channel: FII selling increased supply of Indian equities and reduced demand, directly depressing prices through market microstructure dynamics. The indirect channel: FII selling signalled global risk aversion, which led domestic investors to revise their portfolios defensively, amplifying the initial price impact. Murthy and Deb (2014) estimated that the two channels together explain approximately 60–70% of the Sensex's decline from its January 2008 peak to its March 2009 trough, with the remainder attributable to revisions in corporate earnings expectations driven by the real economic slowdown.

8.2 The 2013 Taper Tantrum

The 2013 'taper tantrum' — triggered by Federal Reserve Chairman Bernanke's May 22, 2013 congressional testimony indicating the Fed might begin tapering its asset purchase programme — provides a cleaner test of the FII transmission mechanism because it was triggered by a single identifiable global event without the simultaneous domestic economic disruption that complicated the GFC analysis.

Sehgal and Tripathi (2015) used an event study methodology to examine the impact of the taper announcement on FII flows and market returns. They found that the announcement was associated with a significant change in FII flow dynamics — from net buying in the weeks before the announcement to net selling in the weeks following — and that this change in FII behaviour was associated with a Sensex decline of approximately 9% over the three months following the announcement. The taper tantrum episode demonstrated that Indian markets were sensitive not just to actual changes in global monetary policy but to expectations of future changes, transmitted through the anticipatory portfolio adjustments of FIIs.

8.3 The COVID-19 shock (2020)

The COVID-19 pandemic produced the most dramatic single-month FII outflow in Indian market history: net FII equity outflows in March 2020 alone reached approximately Rs 61,973 crore, as global institutional investors scrambled to reduce risk exposure in response to the rapidly spreading pandemic. The Sensex fell from approximately 41,700 on January 1, 2020 to approximately 25,600 on March 23, 2020 — a 39% decline in less than three months — before recovering equally dramatically.

The post-COVID FII surge is equally noteworthy. As global central banks implemented unprecedented quantitative easing programmes in 2020, FIIs returned to Indian equities with record enthusiasm: net FII equity inflows in fiscal year 2020–21 were approximately Rs 2,74,059 crore, the highest ever recorded, driving the Sensex to a 68% gain for the year. This episode illustrated both the risk of FII-driven market vulnerability during global shock periods and the recovery potential of FII flows once global risk appetite is restored by policy action.

Table 5 presents a comparative analysis of the three major global shock episodes and their impact on FII flows and Indian market performance.

Table 5: Comparative Analysis Of Global Shock Episodes — FII Flows And Market Impact

Episode	Trigger Event	FII Net Flow (₹ Cr.)	Market Peak/Trough	Sensex Decline (%)	Recovery
GFC 2008–09	Lehman collapse (Sep. 2008)	−₹47,706 (FY09)	20,873 → 8,160	−60.9% peak-trough	Took 2+ years
Taper Tantrum 2013	Bernanke speech (May 2013)	−₹16,800 (net, 3 months)	20,286 → 17,905	−11.8% (3 months)	< 6 months
China Shock 2015–16	China devaluation (Aug. 2015)	−₹14,172 (FY16)	29,080 → 22,494	−22.7% (8 months)	~18 months

COVID Crash 2020	Pandemic declaration (Mar. 2020)	−₹61,973 (Mar-20 alone)	41,700 → 25,600	−38.7% (3 months)	< 6 months
Rate Hike Cycle 2022	US Fed hikes begin (Mar. 2022)	−₹37,631 (FY23)	61,476 → 55,132	−10.3% (9 months)	~12 months

Source: SEBI Monthly Bulletins, BSE India historical data, NSDL FPI database. 'Peak/Trough' refers to closing value of BSE Sensex at pre-episode high and subsequent trough. Recovery = approximate time to return to pre-shock levels.

9. The 2014 FPI Regulatory Transition: Market Quality And Investor Behaviour

The replacement of the FII framework with the FPI framework in 2014 — and the further simplification in 2019 — represents the most significant regulatory change in the governance of foreign portfolio investment in India since the original 1993 opening. The academic literature on the market quality and investor behaviour implications of this regulatory transition is still developing, but several important findings have emerged.

9.1 Changes in investor composition

One of the design objectives of the FPI framework was to attract a more diversified and longer-horizon institutional investor base by simplifying the registration process for low-risk entities (Category I/II investors) while maintaining appropriate scrutiny for higher-risk entities (Category III, now Category II). The available SEBI data suggests that this objective has been partly achieved: the proportion of registered FPIs classified as Category I (sovereign funds, pension funds, central banks) increased from approximately 8% of registered FIIs in 2014 to approximately 18% of Category I FPIs by 2023, though Category I investors still represent a small share of total FPI assets under management.

Das and Sundaram (2018) specifically examined whether the FPI transition changed market sensitivity to foreign flows, using a difference-in-differences approach that compared the FII-market regression coefficient before and after the regulatory change. They found a modest but statistically significant reduction in the regression coefficient in the two years following the transition, which they attributed partly to the composition effect described above and partly to the simultaneous growth of domestic institutional counterbalancing that happened to coincide with the regulatory change.

9.2 Structural break evidence

The Bai-Perron structural break analysis applied to the full 2001–2024 monthly dataset (reported in the accompanying thesis) identifies June 2014 as a statistically significant break point in the regression of Sensex returns on FII/FPI flows. The regression coefficient on FII/FPI flows is 0.0034 in the post-2014 period, compared with 0.0058 in the GFC-Recovery period immediately preceding it — a decline of approximately 41%. This structural break is one of three identified in the full dataset (the others being September 2008 and May 2013), and its location at the time of the FPI regulatory transition provides empirical support for the hypothesis that the regulatory change had a measurable effect on the FII-market relationship.

The interpretation of this structural break is, however, subject to an important caveat: the 2014–2020 period was also a period of rapid growth in DII assets under management, driven by the SIP boom in domestic equity mutual funds. It is therefore not possible from the available data to determine whether the reduction in the FII/FPI-market regression coefficient in this period reflects the regulatory transition per se, the growth of DII counterbalancing, or both. Future research using sub-period data on DII flows alongside FPI flows would help disentangle these two concurrent developments.

10. The Counterbalancing Role Of Domestic Institutional Investors

The growing literature on the FII/FPI-market relationship in India has increasingly recognized that the relationship cannot be understood without simultaneously examining the behaviour of Domestic Institutional Investors (DIIs). DIIs — primarily domestic equity mutual funds, insurance companies, and pension funds — have grown substantially over the study period and have developed a pattern of behaviour that partially counteracts FII activity: buying when FIIs sell and selling (or reducing purchases) when FIIs buy.

10.1 The DII-FII inverse relationship

Multiple studies have documented a negative correlation between FII net equity purchases and DII net equity purchases in the Indian market. Rastogi and Chatterjee (2020) computed the correlation between monthly FII and DII net equity purchases for the period 2015–2019 and found a coefficient of approximately −0.45, indicating a moderate but significant tendency for DIIs to buy when FIIs sell and vice versa. SEBI data for the broader 2014–2024 period confirms and strengthens this finding: the monthly correlation between FII and DII net purchases is approximately −0.52 over this period.

Table 6 provides a comparative overview of FII and DII net equity investment flows in India, illustrating the counterbalancing pattern in specific years.

Table 6: FII vs. DII Net Equity Investment — Counterbalancing Dynamics (₹ Crore)

Fiscal Year	FII/FPI Net (₹ Cr.)	DII Net (₹ Cr.)	FII+DII Combined	Market Outcome
2015–16	−14,172	+65,816	+51,644	DII cushioned FPI outflow; Sensex −9.4%
2018–19	−59,588	+1,27,014	+67,426	Strong DII buying offset FPI; Sensex +17.3%
2020–21 (H1)	−65,000 (est.)	+90,000 (est.)	+25,000 (est.)	COVID crash cushioned by DII
2020–21 (H2)	+3,39,000	+15,000	+3,54,000	FPI + DII drove 68% Sensex gain
2022–23	−37,631	+1,59,485	+1,21,854	DIIs absorbed FPI selling; Sensex −1%
2023–24	+2,06,238	+35,000 (est.)	+2,41,238	FPI return; robust market (+25%)

Source: SEBI Monthly Bulletins, AMFI data, BSE India. H1 and H2 = first and second halves of fiscal year 2020–21. 'est.' = estimated from available monthly data. DII includes domestic mutual funds, insurance companies, and other domestic institutions.

Table 6 illustrates the counterbalancing pattern with particular clarity in fiscal year 2022–23. Despite massive FPI outflows of Rs 37,631 crore driven by the US rate hike cycle and the Russia-Ukraine geopolitical shock, DII net equity purchases of Rs 1,59,485 crore more than compensated for the FPI selling, and the Sensex declined by only 1% for the full fiscal year. Without the DII counterbalancing, the same level of FPI outflow would, based on the pre-2014 regression coefficients, have been expected to produce a market decline of the order of 15–20%.

10.2 Mechanism of DII counterbalancing

The DII counterbalancing dynamic works through a combination of automatic and strategic mechanisms. The automatic mechanism operates through the SIP structure of domestic equity mutual fund investments: SIPs create a steady inflow of cash into domestic equity funds regardless of market conditions, forcing fund managers to deploy this cash into equities even when prices are falling. Since falling prices are precisely when FPI selling is most intense, this automatic deployment mechanism creates a natural counterbalancing tendency.

The strategic mechanism operates through the valuation-sensitivity of long-term domestic institutional investors: as FPI selling drives valuations down to levels that appear attractive on fundamental metrics (P/E, P/B, earnings yield), domestic fund managers are incentivised to increase their equity allocations, creating additional buying pressure that moderates the FPI-driven decline. Insurance companies and pension funds, which manage liabilities with long time horizons, are particularly well positioned to take advantage of the buying opportunities created by FPI-driven market dislocations.

10.3 Implications for market stability

The growth of DII counterbalancing capacity has been the most important structural improvement in Indian market stability over the past decade. It has reduced the market's sensitivity to FPI flows (as documented by the weakening regression coefficient in the post-2014 period) and has created a cushion against the most severe FPI outflow episodes. However, DII counterbalancing is not unlimited: in an extreme event — such as a simultaneous global financial crisis and domestic economic shock of the magnitude of the 2008 GFC — the scale of potential FPI outflows could exceed the capacity of DIIs to absorb, even with the greatly expanded DII asset base of the 2020s. The 2008 GFC, in which net FPI outflows of Rs 47,706 crore occurred in a market with a much smaller DII base, provides a cautionary precedent.

11. Research Gaps and Future Research Agenda

The review of the literature identifies five significant gaps that represent opportunities for future research of high academic and policy value.

11.1 Gap 1: The debt-equity nexus in FPI flows

The overwhelming majority of the existing literature examines FII/FPI equity flows in isolation from FPI debt flows. Yet India has also been a significant destination for FPI investment in government and corporate bonds, and the dynamics of debt flows may differ significantly from equity flows — in their sensitivity to global interest rates, in their relationship with equity market performance, and in their implications for exchange rate

and macroeconomic stability. No comprehensive study has simultaneously modelled FPI equity and debt flows alongside equity and bond market variables in India. This represents a significant gap given the increasing integration of equity and fixed income markets in emerging economies.

11.2 Gap 2: Sector-level FPI impact heterogeneity

All existing Indian studies examine FPI flows in aggregate, treating the entire FPI position in India as a single variable. This aggregation necessarily masks heterogeneity in the FPI-market relationship across sectors. FPI investment in the financial sector (banks, NBFCs) may have a different relationship with sector returns than FPI investment in the IT sector, the pharmaceutical sector, or the infrastructure sector. Given that FPI ownership varies widely across sectors — from over 30% in several large IT companies to below 5% in public sector undertakings — a sector-level analysis would provide much more actionable intelligence for investors and more nuanced evidence for regulators.

11.3 Gap 3: Small-cap and mid-cap index dynamics

Almost every study in the FII-market literature uses the BSE Sensex or NSE Nifty 50 as the market variable. These are large-cap indices that track the 30 or 50 largest listed companies. But FPI investment also affects the mid-cap (Nifty Midcap 150) and small-cap (Nifty Smallcap 250) segments of the market. Given that FPI ownership proportions are typically lower in smaller companies and that these segments are less liquid, the market impact of FPI flows per rupee may be larger in the mid-cap and small-cap segments than in the large-cap indices, with important implications for retail investors who are disproportionately invested in smaller companies.

11.4 Gap 4: The FPI investor heterogeneity problem

The existing literature treats the aggregate FPI population as homogeneous, analysing net FPI flows as a single variable without distinguishing between different categories of FPI investor. Yet the FPI population in India is enormously diverse — including long-horizon sovereign wealth funds and pension funds, medium-horizon actively managed equity funds, short-horizon hedge funds, and index-tracking passive funds — whose motivations, investment horizons, market impact, and sensitivity to global signals differ fundamentally. Data at the category level (Category I vs. Category II under the 2019 regulations) would allow testing of the hypothesis that the regulatory transition has changed the composition of the FPI investor base in ways that have moderated market sensitivity to flows.

11.5 Gap 5: Quantifying DII counterbalancing capacity

While the existence of DII counterbalancing is now well-documented, no study has attempted to quantify its capacity — that is, to estimate the maximum scale of FPI outflows that the current DII system can absorb without severe market dislocation. This is a fundamentally important regulatory question: if India's DII sector can now absorb FPI outflows of up to, say, Rs 1 lakh crore per month before market stability is seriously threatened, the regulatory treatment of FPI investment limits should reflect this capacity. If the capacity is lower, the regulatory implications are more cautious.

Table 7 summarizes the identified research gaps and proposes specific research designs for addressing them.

Table 7: Research Gap Summary And Proposed Future Research

Gap No.	Gap Description	Significance	Proposed Research Design
G1	FPI debt and equity flows jointly modelled	Cross-market contagion; bond-equity interaction	VAR model with FPI equity, FPI debt, Nifty, bond yields, INR/USD; daily/monthly data
G2	Sector-level FPI impact heterogeneity	Actionable for sector rotation strategies	Panel regression of 20+ sector indices on sector-specific FPI ownership changes
G3	Mid-cap and small-cap FPI impact	Retail investor risk; market depth policy	Comparative regression: Nifty vs. Nifty Midcap 150 vs. Nifty Smallcap 250 on FPI flows
G4	FPI investor category heterogeneity	Regulatory design implications of FPI transition	Category-wise FPI data from NSDL; separate correlation/regression by investor type

G5	Quantifying DII counterbalancing capacity	Risk-based FPI investment limit design	Stress test methodology: simulate FPI outflow scenarios; estimate DII absorption at each level

12. Discussion: Synthesis of the Literature

The literature on FII/FPI flows and Indian stock market performance has progressed substantially over the three decades since foreign institutional investors entered Indian markets. The early literature, dominated by descriptive studies and simple correlation analyses, established the basic stylized fact: FII flows and Indian market returns are positively and significantly correlated. The intermediate literature, incorporating more sophisticated time-series econometrics, refined this picture by establishing bi-directional Granger Causality, documenting the asymmetry in the market impact of inflows versus outflows, and identifying the role of global push factors in driving both FII flows and market returns. The recent literature has added the institutional dimension — the growth of DII counterbalancing and the impact of the FPI regulatory transition — to produce a more nuanced and realistic account of how foreign portfolio investment interacts with Indian equity markets.

The cumulative picture that emerges from this review can be summarized in eight propositions. First, FII/FPI equity flows and Indian stock market returns are positively and significantly correlated, with the correlation coefficient consistently estimated in the 0.47–0.67 range across multiple studies and time periods. Second, the relationship is characterised by bi-directional Granger Causality at monthly frequency, with the market-to-FII direction somewhat stronger than the FII-to-market direction, consistent with a momentum-following component in FII behaviour. Third, the relationship is asymmetric: FII/FPI outflows produce a 40–50% larger market impact per rupee than equivalent inflows, consistent with fire-sale dynamics and the information signal conveyed by FII selling. Fourth, the relationship is time-varying: it strengthened during the GFC-Recovery period (2008–2014), when FII activity was the dominant driver of market direction, and has weakened in the FPI era (post-2014), as DII counterbalancing capacity has grown. Fifth, the relationship is subject to regime changes associated with global shocks (2008, 2013) and regulatory transitions (2014), which the structural break literature has now identified and quantified.

Sixth, global push factors — particularly global monetary policy (US Fed rate), global risk sentiment (VIX, MSCI EM), and global liquidity conditions — are important determinants of FII flows and must be controlled for in any regression analysis of the FII-market relationship. Studies that do not include adequate global controls may overstate the independent market impact of FII flows. Seventh, the growth of DII counterbalancing has been the most important structural improvement in Indian market stability over the past

decade, moderating the market impact of FPI flows and reducing the risk of a repeat of the extreme 2008-style market decline. Eighth, the existing literature has significant gaps — particularly the absence of debt-equity joint analysis, sector-level analysis, and FPI investor heterogeneity analysis — that future research should address.

13. Conclusion

This review has undertaken a comprehensive and critically organized survey of the academic, regulatory, and empirical literature on the relationship between Foreign Institutional Investment flows and Indian stock market performance, covering 87 studies published between 1993 and 2024. The review has been organized around eight analytical themes — regulatory evolution, theoretical frameworks, correlation and regression evidence, Granger Causality, asymmetry, global shock transmission, the 2014 FPI regulatory transition, and DII counterbalancing — providing, for each theme, a critical evaluation of methodological approaches, key findings, conflicts between studies, and directions for future research.

The review concludes that the FII-market relationship in India is a real, significant, and economically important phenomenon that has been credibly established by a large and methodologically diverse body of empirical research. The basic directional relationship — FII buying is associated with market appreciation; FII selling is associated with market decline — is supported by three decades of evidence and is not seriously contested in the literature. The more nuanced features of the relationship — its bi-directional causal structure, its asymmetry between inflows and outflows, its time-variation across market regimes, and its regime-dependence on regulatory and institutional context — are also well-supported by the evidence, and understanding them is essential for any serious analytical or practical engagement with FII/FPI flows in Indian markets.

For investors, the practical implication is clear: FII flow data is a useful market indicator but not an infallible one, and the information content of FII selling (which has a larger market impact than equivalent buying) deserves particular attention in portfolio risk management. For regulators, the review supports several specific conclusions: that regulatory design matters for market stability outcomes, as evidenced by the structural break associated with the 2014 FPI transition; that the growth of DII counterbalancing capacity deserves active support through financial market development policies; and that the asymmetry of the FII market impact has implications for the calibration of circuit breaker and investment limit rules that may not be fully reflected in current regulatory frameworks.

For researchers, the review has identified five specific gaps — debt-equity joint modelling, sector-level analysis, mid-cap and small-cap impact, FPI investor heterogeneity, and DII counterbalancing capacity quantification — that represent high-priority directions for future work. Progress on these questions would make a genuinely new contribution to the literature and would provide the kind of granular, practically actionable evidence that investors and regulators need to navigate the complex and evolving relationship between foreign portfolio investment and Indian capital markets.

India's capital markets are among the most dynamic and sophisticated in the emerging world, and the question of how they interact with the global capital allocation system through the FPI mechanism is one of the most important and enduring questions in Indian financial economics. This review has attempted to contribute to the ongoing collective effort to answer that question with the rigor, comprehensiveness, and critical clarity that its practical importance demands.

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